

Draft Standard for Local and metropolitan area networks— Link Aggregation Amendment 1: YANG for Link Aggregation

Prepared by the
Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

Sponsor

**LAN/MAN Standards Committee
of the
IEEE Computer Society**

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The text proper of this draft begins with the [Title page](#).

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PAR (Project Authorization Request) and CSD

This page is a draft, based on the proposed PAR and CSD as of the close of the May 2023 802.1 Interim Meeting.

Extracts from the PAR, as approved by IEEE NesCom February 15, 2024:

<https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/10885>

and the CSD (Criteria for Standards Development):

<https://mentor.ieee.org/802-ec/dcn/23/ec-23-0238-00-ACSD-p802-1axdz.pdf>

follow.

Scope of the project:

This amendment specifies YANG modules that allows configuration and status reporting for systems implementing Link Aggregation, and optionally Distributed Resilient Network Interconnect, based on the capabilities currently specified in clause 7 (management) and Annex D (Management Information Base definitions). This amendment also includes technical and editorial corrections in the description of existing IEEE Std 802.1AX functionality.

PAR Need for the Project:

YANG (IETF RFC 7950) is a formalized data modeling language that is widely accepted and can be used to simplify network configuration. The ability to manage Link Aggregation via YANG modules is needed for compatibility with modern network management systems.

CSD broad market potential [extract]:

The proposed amendment will support the use of YANG, which has broad industry support in networks that use IEEE Std 802.1AX. Both IEEE Std 802.1AX and YANG are already supported and used by multiple vendors, network providers, and network users. There is a wide interest in the industry to manage Link Aggregation via YANG.

Economic feasibility [extract]:

- a) Management using YANG utilizes a balance between end station and infrastructure capabilities; the balance will be similar to that for existing management methods.
- b) The cost factors will be similar to those of existing management methods.
- c) This project adds YANG capabilities to IEEE Std 802.1AX as a step towards a complete YANG management solution. This helps to eliminate multiple management platforms, thus reduces installation cost.
- d) This project adds YANG capabilities to IEEE Std 802.1AX as a step towards a complete YANG management solution. This helps to eliminate multiple management platforms, thus reduces operational cost.

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MIB and YANG modules

The MIB and YANG modules that are modified or added by this amendment are attached to the draft pdf as plain text (UTF-8) .mib and .yang files. When a roll up of the current base standard plus this amendment is made available, all the MIB and YANG modules for the roll up are attached.

Introduction to the current draft ¹

This introduction is not part of the draft, and should not be the subject of ballot comments.

D1.2

Incorporates the resolution of comments received during the Working Group recirculation ballot of draft 1.1. Significant changes resulting from the ballot include:

- Incorporate changes suggested by the IEEE MEC review.
- Update the regex pattern for strings in a drni-mask grouping.
- Correct the namespace to urn:ieee:std:802.1AX:yang:ieee802-dot1ax-<types/linkagg/drni>.

D1.1

Incorporates the resolution of comments received during the Working Group ballot of draft 1.0. Significant changes resulting from the ballot include:

- Changing the representation of the 4096 bit mask in the drni-mask grouping from binary to a string listing individual CIDs, or ranges of CIDs, corresponding to bits in the mask that are set to 1.
- Removing several 4096 bit masks from the YANG model because they could be calculated from other information in the model;
- Moving the actor-admin-state leaf from the aggport container to the key-group container.

D1.0

Incorporates the resolution of comments received during the Task Group ballot of draft 0.2. Significant changes resulting from the ballot include:

- Combining the old figures 10-2 and 10-3 into a single page UML-like diagram for Link Aggregation (excluding DRNI);
- Clean up separation of Link Aggregation from the optional DRNI by adding a separate line in the conformance clause options for DRNI, and moving all DRNI related types from *ieee802-dot1ax-types.yang* to *ieee802-dot1ax-drni.yang*;
- Moving the system and system-priority leafs from key-groups to a separate list of aggregation systems. This allows the system mac address to be provided by the system rather than requiring it to be explicitly specified in a configuration model.

Also changed was adding a base identity choice for the admin-conv-link-map and admin-conv-service-map of CSCD. This allows a vendor to augment the yang model with an identity statement specifying pre-defined contents of the table to provide a shorthand for configuration that otherwise could potentially take thousands of lines of configuration.

D0.2

Incorporates the resolution of comments received during the Task Group ballot of draft 0.1.

¹ The whole or parts of the introduction, possibly updated, to past drafts may be retained at the Editor's discretion, with the most recent introduction first. The introduction to each draft may solicit input on specific subjects.

D0.1

This is an initial draft and comments are requested on all aspects of the draft. It includes a number of notes that may be of help to the Editors as well as informing the initial review process.

Stephen Haddock, 802.1AXdz Editor

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Abstract: This amendment to IEEE Std 802.1AX-2020 specifies a Unified Modeling Language (UML)-based model and YANG modules for Link Aggregation configuration and status reporting.

Keywords: Aggregated Link, Aggregator, Distributed Resilient Network Interconnect, DRNI, interconnect, Link Aggregation, Link Aggregation Group, local area network, management, Network-Network Interface, NNI, YANG.

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The following members of the individual balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

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When the IEEE-SA Standards Board approved this standard on XX Month 20xx, it had the following membership:

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Introduction

This introduction is not part of IEEE Std 802.1AXdz™-20XX, IEEE Standard for Local and metropolitan area networks—Link Aggregation—Amendment 1: YANG for Link Aggregation.

IEEE Std 802.1AXdz™-202X: YANG for Link Aggregation specifies a Unified Modeling Language (UML)-based model and YANG modules for Link Aggregation configuration and status reporting

This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution. Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and to incorporate new related material. Information on the current revision state of this and other IEEE 802 standards may be obtained from

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2. Normative references

Insert the following references into Clause 2 in alphanumeric order:

IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011.

IETF RFC 6991, Common YANG Data Types, July 2013

IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

IETF RFC 8343, A YANG Data Model for Interface Management, March 2018

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4. Acronyms and abbreviations

Insert the following acronyms into Clause 4 in alphabetical order:

NETCONF Network Configuration Protocol

UML® Unified Modeling Language™¹

¹ UML® is a registered trademark of the Object Management Group, Inc., and Unified Modeling Language™ is a trademark of the Object Management Group, Inc.

5. Conformance

Insert the following text (items j and k) after item i) in the lettered list of 5.3.2

5.3.2 Link Aggregation options

- j) Support YANG modules for the management of Link Aggregation capabilities (Clause 10).
- k) Support YANG modules for the management of DRNI capabilities (Clause 10).

Insert Clause 10 after Clause 9 as follows:

10. YANG Data Model

This clause specifies the YANG (IETF RFC 7950) data model that provides control and status monitoring of systems and system components that implement functionality specified in this standard.

This clause:

- a) Introduces the YANG framework that governs the naming and hierarchy of configuration and operational data structures in the data models, and the modeling of network interfaces (10.1).
- b) Describes the information data model and its relationship to the operational processes and managed objects specified in the other clauses of this standard, and provides a representation, similar to the Unified Modeling Language (UML), of relevant objects (10.2).
- c) Describes the structure of the YANG data model, which comprises three YANG modules (10.3).
- d) Includes a relationship description of other modules imported in YANG modules (10.4)
- e) Reviews security considerations applicable to each of the modules, with specific reference to data nodes in the YANG modules that compose the YANG data model (10.5).
- f) Includes each of the YANG modules and its data schema (10.6).

10.1 YANG Framework

The YANG framework applies hierarchy in the following areas:

- a) The uniform resource name (URN), as specified in 802d.
- b) The YANG objects form a hierarchy of configuration and operational data structures that specify the YANG model.

10.2 Information Model for Link Aggregation Management

The YANG objects are based on the managed objects in Clause 7. A UML-like representation of the management model is provided in the following subclauses.

NOTE—OMG® UML2.5² [B11] conventions together with C++ language constructs are used in this clause as a representation to convey model structure and relationships..

The purpose of an UML-like³ diagram is to express the model design on a single piece of paper. The structure of the UML-like representation shows the name of the object followed by a list of properties for the object. The properties indicate its type and accessibility. The UML-like representation is meant to express simplified semantics for the properties. It is not meant to provide the specific datatype as used to encode the object in either MIB or YANG. In the UML-like representation, a box with a white background represents information that comes from sources outside of the IEEE. A box with a gray background represents objects that are specified by this IEEE Standard.

The YANG hierarchical structure that incorporates the Link Aggregation YANG modules supported by this standard is represented by Figure 10-1. In the figures in this clause, items that are shaded gray are described in this document, items with no background shading are specified elsewhere. The YANG data model is

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³A description of the UML-like diagrams used in this clause is provided at <https://1.ieee802.org/uml-like-diagrams>.

realized in three YANG modules. One module *ieee802-dot1ax-types* provides data types that are needed by the Link Aggregation configuration and monitoring objects. The *ieee802-dot1ax-linkagg* module provides the Link Aggregation, Aggregation Port, and Aggregator configuration and monitoring objects. The *ieee802-dot1ax-drni* module provides the Distributed Resilient Network Interface (DRNI) configuration and monitoring objects. The Link Aggregation and DRNI capabilities are not only applicable to IEEE Std 802.1Q bridges, but also (for example) end stations, and routers.

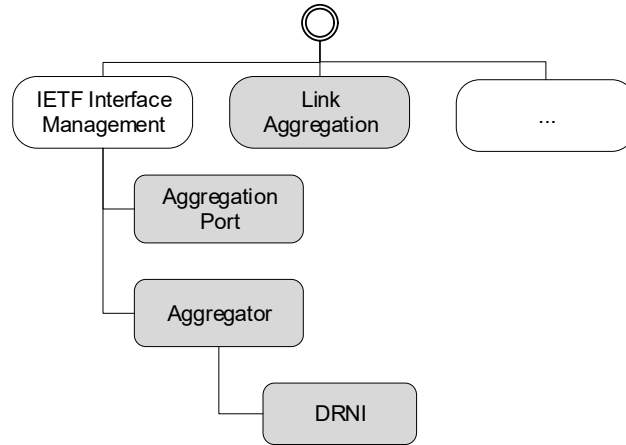


Figure 10-1—YANG root hierarchy with Link Aggregation YANG modules

10.2.1 Link Aggregation model

The Link Aggregation Configuration and Monitoring Objects in Figure 10-2 show the objects that are applicable on a system supporting link aggregation, and the interfaces supporting link aggregation.

The objects on the system consist of a list of aggregation systems, and a list of key groups. A key group is the set of aggregators and aggregation ports that share the same system priority, system mac address, and aggregation key, and therefore can potentially form a Link Aggregation Group. Each key group has a unique combination of actor-admin-key and actor-system, and includes parameters that have the same value for any aggregator and/or aggregation port that have the same actor-admin-key and actor-system.

NOTE—See 6.3.5 and 6.3.6 for a description of the identification of Aggregation Ports that can potentially form a Link Aggregation Group, and Figure 6-15 for examples.

Each Aggregation Port is an interface (e.g. of type ethernetCsmacd) augmented with an “aggport” presence container as shown on the right side of Figure 10-2. Each Aggregator is an interface to the entire Link Aggregation Group (using type *ieee8023adlag*⁴) augmented with a “lag” container as shown on the left side of Figure 10-2. The binding of Aggregation Ports to Aggregators is a dynamic function of the Link Aggregation Control Protocol. The Aggregation Ports attached to an Aggregator can be read from the *lower-layer-if* attribute of the Aggregator interface. The Aggregator to which an Aggregation Port is attached can be read through the *higher-layer-if* attribute of the Aggregation Port interface. An Aggregation Port will only transmit and respond to LACPDU's when the interface “enabled” leaf is true, and will only attach to an Aggregator when the Aggregator “enabled” leaf is true. The interface “oper-status” leaf of an Aggregation Port will be “up” when the Aggregation Port is attached to an Aggregator and the “collecting” and “distributing” bits of the “actor-oper-state” leaf are both true. The interface “oper-status” leaf on an Aggregator will be “up” when it has at least one attached Aggregation Port that is “up”.

⁴Link Aggregation was originally published as IEEE Std 802.3ad-2000. It was moved to the 802.1 Working Group in 2008 and subsequent revisions have been published as IEEE Std 802.1AX.

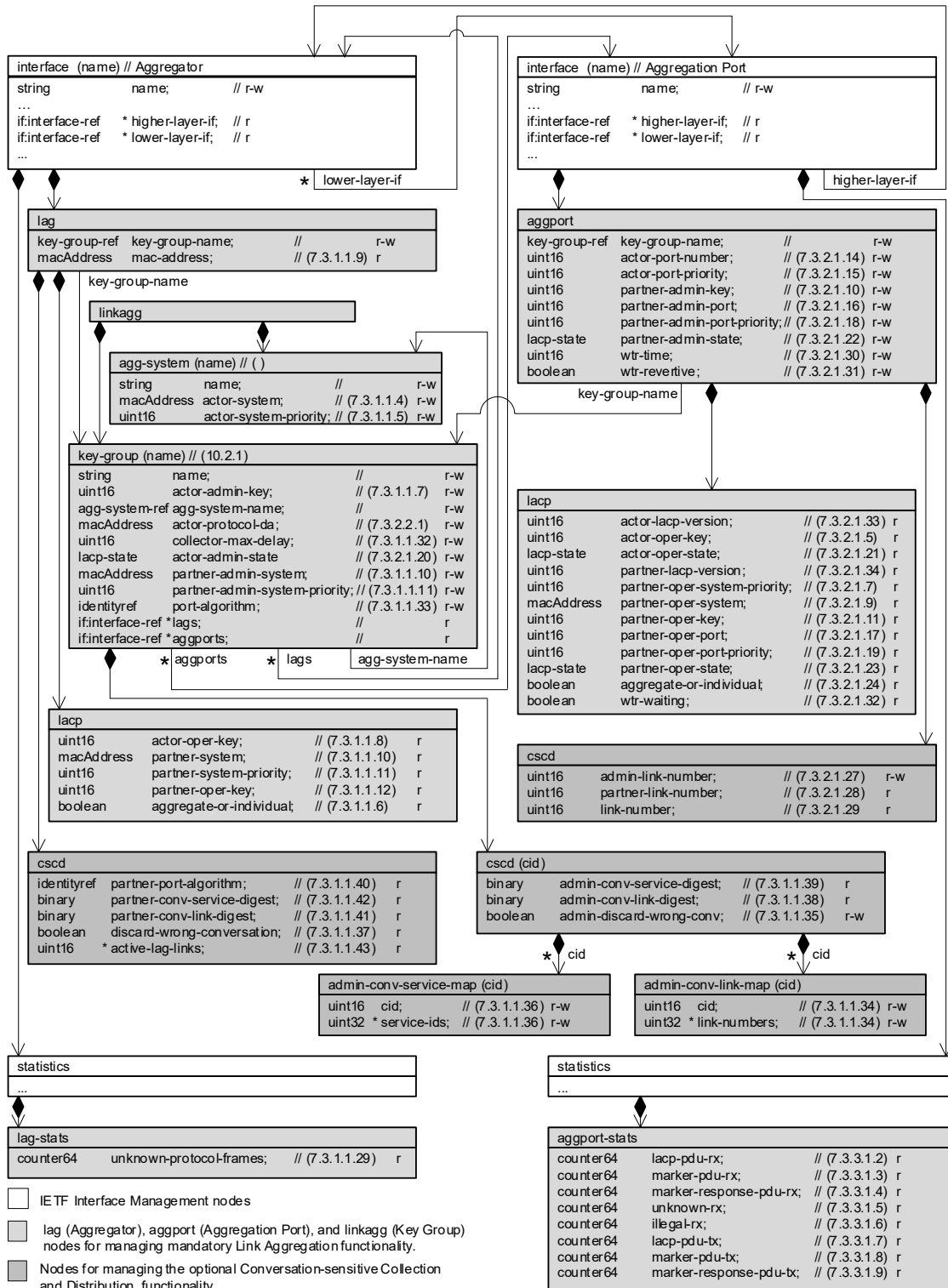


Figure 10-2—Link Aggregation Configuration and Monitoring Objects

10.2.2 DRNI model

The DRNI Configuration and Monitoring Objects in Figure 10-3 show the objects that are applicable to an Aggregator augmented with a “drni” container.

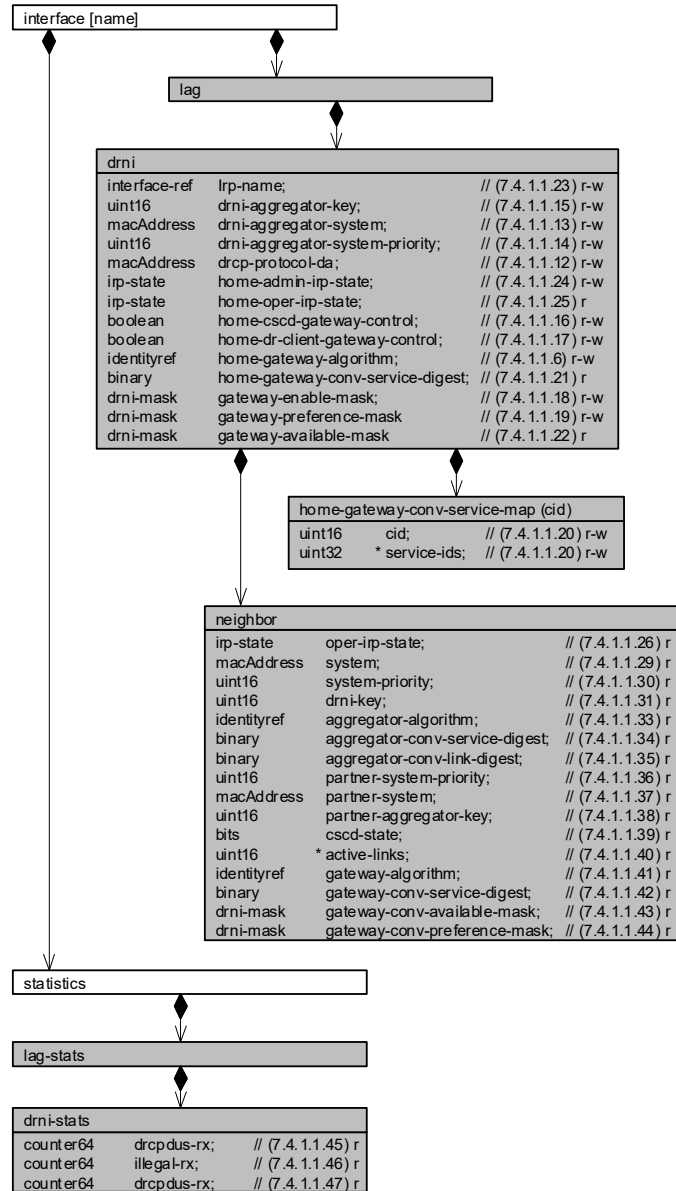


Figure 10-3—DRNI Configuration and Monitoring Objects

10.3 Structure of the Link Aggregation YANG Model

The YANG data model specified in this standard is comprised of three YANG modules. A summary of the modules contained in this clause is represented in Table 10-1.

Table 10-1—Structure of the YANG modules

Module	Subclause	Notes
ieee802-dot1ax-types	10.7.1	Type definitions used for Link Aggregation YANG.
ieee802-dot1ax-linkagg	10.7.2	Link Aggregation Management
ieee802-dot1ax-drni	10.7.3	DRNI Management

In the YANG modules below, if any discrepancy between the DESCRIPTION text and the corresponding specification in any other part of this standard occurs, the specifications outside this clause take precedence.

10.4 Relationship to other YANG modules

This clause describes how the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules are related to the YANG modules that are imported.

10.4.1 IEEE 802.1AX Types Module

The *ieee802-dot1ax-types* module provides reusable types that are used by the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* modules.

10.4.2 IETF YANG Types Module

The *ietf-yang-types* YANG module (IETF RFC 6991) contains a set of derived YANG types. This document leverages counter64.

10.4.3 IETF Interfaces YANG Module

The *ietf-interfaces* YANG module (IETF RFC 8343) contains a set of YANG definitions for managing network interfaces. This document augments an *ietf-interfaces:interface* with aggregation-port or aggregator data nodes..

10.4.4 IEEE 802 Types Module

The *ieee802-types* module provides reusable types that are used in IEEE 802 standards.

NOTE—The type for mac-addresses defined in *ieee802-types* has a pattern that allows upper and lower case letters. To avoid issues with string comparison, it is suggested to only use upper case for the letters in the hexadecimal numbers. Implementers using code comparing MAC addresses are advised that there is still an issue with a difference between the IETF mac-address definition and the IEEE mac-address definition.

10.5 Security Considerations

The YANG modules specified in this clause are designed to be accessed via a network configuration protocol (e.g., NETCONF protocol). In the case of NETCONF, the lowest NETCONF layer is the secure transport layer and the mandatory to implement secure transport is SSH. The NETCONF access control model provides the means to restrict access for particular NETCONF users to a pre-configured subset of all available NETCONF protocol operations and content.

It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those users who have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

10.5.1 Security considerations of the *ieee802-dot1ax* YANG modules

There are several management objects specified in the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules that are configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects could be considered sensitive or vulnerable in some network environments. A network configuration protocol, such as NETCONF (IETF RFC 6241), can support protocol operations that can edit or delete YANG module configuration data (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure environment without proper protection, then negative effects on the network operation are possible.

The following containers, and the objects in these containers, of the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules can be manipulated to interfere with the operation of the Link Aggregation Control Protocol (LACP) or the Distributed Relay Control Protocol (DRCP). This could, for example, cause network instability and result in the loss of service for a large number of end users.

```
dot1ax-linkagg/linkagg/key-group
dot1ax-linkagg/linkagg/key-group/cscd
dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-service-map
dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-link-map
dot1ax-linkagg/aggport
dot1ax-linkagg/aggport/cscd/admin-link-number
dot1ax-linkagg/lag
dot1ax-drni/lag/drni
dot1ax-drni/lag/drni/home-gateway-conv-service-map
```

Some of the readable data in this YANG module could be considered sensitive or vulnerable in some network environments. It is important to control all types of access (e.g., including NETCONF get, get-config operations) to these objects and possibly to even encrypt the values of these objects when sending them over the network. For example, the system name and other information about the remote systems could provide information about the configuration and topology of the network and could be considered a privacy threat.

10.6 YANG schema tree definitions^{5,6}

A simplified graphical representation of the data model is used in this document. The meaning of the symbols in these diagrams is as follows:

- Brackets "[" and "]" enclose list keys.
- Abbreviations before data node names: "rw" means configuration (read-write), and "ro" means state data (read-only).
- Symbols after data node names: "?" means an optional node, "!" means a presence container, and "*" denotes a list and leaf-list.
- Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (":").
- Ellipsis ("...") stand for contents of subtrees that are not shown.

10.6.1 Schema for the *ieee802-dot1ax-linkagg* YANG module

```

module: ieee802-dot1ax-linkagg
+--rw linkagg
  +--rw agg-system* [name]
    | +--rw name string
    | +--rw actor-system? ieee:mac-address
    | +--rw actor-system-priority? uint16
  +--rw key-group* [name]
    +--rw name string
    +--rw actor-admin-key uint16
    +--rw agg-system-name agg-system-ref
    +--rw actor-protocol-da? ax:bridge-reserved-addresses
    +--rw collector-max-delay? uint16
    +--rw actor-admin-state? ax:lacp-state
    +--rw partner-admin-system? ieee:mac-address
    +--rw partner-admin-system-priority? uint16
    +--rw port-algorithm? identityref
    +--ro lags* if:interface-ref
    +--ro aggports* if:interface-ref
    +--rw cscd {ax:cscd}?
      +--rw admin-conv-service-map {ax:sid-map}?
        | +--rw (method)?
        | | +--:(pattern)
        | | | +--rw pattern? identityref
        | | +--:(cid-list)
        | | | +--rw cid-list* [cid]
        | | | | +--rw cid uint16
        | | | | +--rw service-ids* uint32
        +--ro admin-conv-service-digest? binary
      +--rw admin-conv-link-map
        | +--rw (method)?
        | | +--:(pattern)
        | | | +--rw pattern? identityref
        | | +--:(cid-list)
        | | | +--rw cid-list* [cid]
        | | | | +--rw cid uint16
        | | | | +--rw link-numbers* uint16
        +--ro admin-conv-link-digest? binary
      +--rw admin-discard-wrong-conv? enumeration {ax:dwc}?

augment /if:interfaces/if:interface:
+--rw lag
  +--rw key-group-name key-group-ref
  +--ro mac-address? ieee:mac-address
  +--ro lacp
  | +--ro actor-oper-key? uint16

```

⁵Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

⁶An ASCII version of the YANG module(s) can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

```

| +--ro partner-system?          ieee:mac-address
| +--ro partner-system-priority? uint16
| +--ro partner-oper-key?        uint16
| +--ro aggregate-or-individual? boolean
+--ro cscd {ax:cscd}?
  +--ro partner-port-algorithm?   identityref
  +--ro partner-conv-service-digest? binary
  +--ro partner-conv-link-digest? binary
  +--ro discard-wrong-conversation? boolean
  +--ro active-lag-links*         uint16
augment /if:interfaces/if:interface/if:statistics:
  +--ro lag-stats
  +--ro unknown-protocol-frames? yang:counter64
augment /if:interfaces/if:interface:
  +--rw aggport!
  +--rw key-group-name            key-group-ref
  +--rw actor-port-number?        uint16
  +--rw actor-port-priority?      uint16
  +--rw partner-admin-key?        uint16
  +--rw partner-admin-port?       uint16
  +--rw partner-admin-port-priority? uint16
  +--rw partner-admin-state?      ax:lacp-state
  +--rw wtr-time?                 uint16
  +--rw wtr-revertive?            boolean
  +--ro lacp
  | +--ro actor-lacp-version?      uint8
  | +--ro actor-oper-key?          uint16
  | +--ro actor-oper-state?        ax:lacp-state
  | +--ro partner-lacp-version?    uint8
  | +--ro partner-oper-system-priority? uint16
  | +--ro partner-oper-system?     ieee:mac-address
  | +--ro partner-oper-key?        uint16
  | +--ro partner-oper-port?       uint16
  | +--ro partner-oper-port-priority? uint16
  | +--ro partner-oper-state?      ax:lacp-state
  | +--ro aggregate-or-individual? boolean
  | +--ro wtr-waiting?             boolean
  +--rw cscd {ax:cscd}?
    +--rw admin-link-number?      uint16
    +--ro partner-link-number?    uint16
    +--ro link-number?            uint16
augment /if:interfaces/if:interface/if:statistics:
  +--ro aggport-stats
  +--ro lacp-pdu-rx?               yang:counter64
  +--ro marker-pdu-rx?             yang:counter64
  +--ro marker-response-pdu-rx?    yang:counter64
  +--ro unknown-rx?                yang:counter64
  +--ro illegal-rx?                yang:counter64
  +--ro lacp-pdu-tx?               yang:counter64
  +--ro marker-pdu-tx?             yang:counter64
  +--ro marker-response-pdu-tx?    yang:counter64

```

10.6.2 Schema for the *ieee802-dot1ax-drni* YANG module

```

module: ieee802-dot1ax-drni

augment /if:interfaces/if:interface/dot1ax:lag:
  +--rw drni!
  +--rw irp-name                  if:interface-ref
  +--rw drni-aggregator-key?      uint16
  +--rw drni-aggregator-system?   ieee:mac-address
  +--rw drni-aggregator-system-priority? uint16
  +--rw drcp-protocol-da?         ax:bridge-reserved-addresses
  +--rw home-admin-irp-state?     irp-state
  +--ro home-oper-irp-state?      irp-state
  +--rw home-cscd-gateway-control? boolean
  +--rw home-dr-client-gateway-control? boolean
  +--rw home-gateway-algorithm?   identityref
  +--rw home-gateway-conv-service-map {ax:sid-map}?
  | +--rw (method)?

```

```

|      +---:(pattern)
|      | +--rw pattern?      identityref
|      +---:(cid-list)
|      |   +--rw cid-list* [cid]
|      |   +--rw cid          uint16
|      |   +--rw service-ids* uint32
+--ro home-gateway-conv-service-digest?  binary
+--rw gateway-enable-mask
| +--rw cid-str*      string
| +--rw invert-mask?  boolean
+--rw gateway-preference-mask
| +--rw cid-str*      string
| +--rw invert-mask?  boolean
+--ro gateway-available-mask
| +--ro cid-str*      string
| +--ro invert-mask?  boolean
+--ro neighbor
  +--ro oper-irp-state?      irp-state
  +--ro system?              ieee:mac-address
  +--ro system-priority?     uint16
  +--ro drni-key?            uint16
  +--ro aggregator-algorithm? identityref
  +--ro aggregator-conv-service-digest?  binary
  +--ro aggregator-conv-link-digest?     binary
  +--ro partner-system-priority?          uint16
  +--ro partner-system?                   ieee:mac-address
  +--ro partner-aggregator-key?           uint16
  +--ro cscd-state?                       bits
  +--ro active-links*                     uint16
  +--ro gateway-algorithm?                identityref
  +--ro gateway-conv-service-digest?      binary
  +--ro gateway-available-mask
  | +--ro cid-str*      string
  | +--ro invert-mask?  boolean
  +--ro gateway-preference-mask
  | +--ro cid-str*      string
  | +--ro invert-mask?  boolean
augment /if:interfaces/if:interface/if:statistics/dot1ax:lag-stats:
+--ro drni-stats
  +--ro drcpdus-rx?  yang:counter64
  +--ro illegal-rx?  yang:counter64
  +--ro drcpdus-tx?  yang:counter64

```

10.7 YANG modules

10.7.1 The *ieee802-dot1ax-types* YANG module

```

module ieee802-dot1ax-types {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-types";
  prefix dot1ax-types;

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://www.ieee802.org/1/
    WG-Email: stds-802-1-1@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
            IEEE Standards Association
            45 Hoes Lane
            Piscataway, NJ 08854
            USA
    E-mail: stds-802-1-chairs@ieee.org";
  description
    "Common types used within 802.1AX Link Aggregation modules."

```

Copyright (C) IEEE (2025).

This version of this YANG module is part of IEEE Std 802.1AX;
see the standard itself for full legal notices.";

```
revision 2025-03-11 {
  description
    "Published as part of IEEE Std 802.1AXdz-202y.
    The following reference statement identifies each referenced IEEE
    Standard as updated by applicable amendments.";
  reference
    "IEEE Std 802.1AX-2020 Link Aggregation:
    IEEE Std 802.1AXdz-202y.";
}

feature cscd {
  description
    "Conversation Sensitive Collection and Distribution (CSCD)
    is supported.";
  reference
    "5.3.2, 6.6 of IEEE Std 802.1AX";
}

feature dwc {
  description
    "The Discard Wrong Conversation option in CSCD is
    supported.";
  reference
    "5.3.2, 6.6 of IEEE Std 802.1AX";
}

feature sid-map {
  description
    "The Service ID to Conversation ID map is supported for
    use by distribution algorithms.";
}

feature c-vid-dist-alg {
  description
    "Supports distribution based on C-VIDs.";
}

feature s-vid-dist-alg {
  description
    "Supports distribution based on S-VIDs.";
}

feature i-sid-dist-alg {
  description
    "Supports distribution based on I-SIDs.";
}

feature te-sid-dist-alg {
  description
    "Supports distribution based on TE-SIDs.";
}

feature flow-hash-dist-alg {
  description
    "Supports distribution based on Flow Hash.";
}

feature basic-link-maps {
  description
    "Pre-defined admin-conv-link-map configurations to distribute
    packets on either one or two active links, with any
    remaining links available as standby.";
}

identity distribution-algorithm {
  description
    "Each distribution algorithm is identified by a sequence of
```

4 octets, structured as shown in Figure 8-1. Distribution algorithm identifiers are used by network administrators to select between algorithms and, in Conversation-sensitive LACP and (if supported) Distributed Resilient Network Interconnect (DRNI) operation, to check whether partners and neighbors are using the same algorithm.

This identity is intended to serve as base identity, not to be directly referenced.

Vendor specific, combination (ex: multi-layer), and other customized distribution algorithms can be created as their own identities in their own YANG files, derived from this imported base type. When the identity description specifies a 4 octet value, this value is used for cscd purposes, and included in LACPDUs. Otherwise the identity is used purely for local selection of a distribution algorithm, and the unspecified distribution algorithm is used for cscd purposes.";

```
reference
  "8.1, 8.2 of IEEE Std 802.1AX";
}

identity unspecified {
  base distribution-algorithm;
  description
    "The 'Unspecified distribution algorithm' identifier has
    been reserved for use when the algorithm is unknown (or
    is not advertised).
    The algorithm identifier is 00-80-C2-00.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity c-vids-nomap {
  if-feature "c-vid-dist-alg";
  base distribution-algorithm;
  description
    "Distribution based on C-VIDs (8.2.1). No Service ID
    mapping table is used.
    The algorithm identifier is 00-80-C2-01.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity c-vids-map {
  if-feature "c-vid-dist-alg and sid-map";
  base distribution-algorithm;
  description
    "Distribution based on C-VIDs (8.2.1). A Service ID
    mapping table is used.
    The algorithm identifier is 00-80-C2-81.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity s-vids-nomap {
  if-feature "s-vid-dist-alg";
  base distribution-algorithm;
  description
    "Distribution based on S-VIDs (8.2.2). No Service ID
    mapping table is used.
    The algorithm identifier is 00-80-C2-02.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity s-vids-map {
  if-feature "s-vid-dist-alg and sid-map";
  base distribution-algorithm;
  description
    "Distribution based on S-VIDs (8.2.2). A Service ID
```



```
        mapping table is used.
        The algorithm identifier is 00-80-C2-82.";
    reference
        "Table 8-1 of IEEE Std 802.1AX";
}

identity i-sids-nomap {
    if-feature "i-sid-dist-alg";
    base distribution-algorithm;
    description
        "Distribution based on I-SIDs (8.2.3). No Service ID
        mapping table is used.
        The algorithm identifier is 00-80-C2-03.";
    reference
        "Table 8-1 of IEEE Std 802.1AX";
}

identity i-sids-map {
    if-feature "i-sid-dist-alg and sid-map";
    base distribution-algorithm;
    description
        "Distribution based on I-SIDs (8.2.3). A Service ID
        mapping table is used.
        The algorithm identifier is 00-80-C2-83.";
    reference
        "Table 8-1 of IEEE Std 802.1AX";
}

identity te-sids-nomap {
    if-feature "te-sid-dist-alg";
    base distribution-algorithm;
    description
        "Distribution based on TE-SIDs (8.2.4). No Service ID
        mapping table is used.
        The algorithm identifier is 00-80-C2-04.";
    reference
        "Table 8-1 of IEEE Std 802.1AX";
}

identity te-sids-map {
    if-feature "te-sid-dist-alg and sid-map";
    base distribution-algorithm;
    description
        "Distribution based on TE-SIDs (8.2.4). A Service ID
        mapping table is used.
        The algorithm identifier is 00-80-C2-84.";
    reference
        "Table 8-1 of IEEE Std 802.1AX";
}

identity flow-hash-nomap {
    if-feature "flow-hash-dist-alg";
    base distribution-algorithm;
    description
        "Distribution based on Flow Hash (8.2.5). No Service ID
        mapping table is used.
        The algorithm identifier is 00-80-C2-05.";
    reference
        "Table 8-1 of IEEE Std 802.1AX";
}

identity flow-hash-map {
    if-feature "flow-hash-dist-alg and sid-map";
    base distribution-algorithm;
    description
        "Distribution based on Flow Hash (8.2.5). A Service ID
        mapping table is used.
        The algorithm identifier is 00-80-C2-85.";
    reference
        "Table 8-1 of IEEE Std 802.1AX";
}
```

```
identity link-map-patterns {
  description
    "Base identity for patterns filling the admin-conv-link-map.
    This identity is intended to serve as base identity, not
    to be directly referenced. Use of the identity allows vendors
    to augment the module with vendor specified patterns,";
  reference
    "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
}

identity one-plus-n {
  if-feature "basic-link-maps";
  base link-map-patterns;
  description
    "Provides active/standby behavior with one active link and
    up to 63 standby links. All packets are mapped to whichever
    link that is active in the aggregation has the lowest link
    number. If that link fails, all packets are mapped to the
    lowest link number of the remaining links that are active
    in the aggregation.
    The map consists of an identical entry for each CID, with a
    list of numbers from 1 to 64.
    Supports any number of links in the aggregation (up to 64)
    with link numbers between 1 and 64. Any links in the
    aggregation with a link number greater than 64 are not
    used.";
}

identity two-plus-n {
  if-feature "basic-link-maps";
  base link-map-patterns;
  description
    "Provides basic load-balancing on two active links with up
    to 62 standby links. All packets with an even CID value are
    mapped to the link with the lowest link number in the
    aggregation, and all packets with an odd CID value are
    mapped to the link with the highest link number in the
    aggregation.
    The map consists of an identical entry for each even CID
    containing a list of numbers from 1 to 64, and an identical
    entry for each odd CID containing a list of numbers from 64
    to 1.
    Supports any number of links in the aggregation (up to 64)
    with link numbers between 1 and 64. Any links in the
    aggregation with a link number greater than 64 are not
    used.";
}

identity service-map-patterns {
  description
    "Base identity for patterns filling the admin-conv-service-map.
    This identity is intended to serve as base identity, not
    to be directly referenced. Use of the identity allows vendors
    to augment the module with vendor specified patterns,";
  reference
    "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
}

typedef lacp-state {
  type bits {
    bit lacp-activity {
      position 1;
      description
        "Provides administrative control over when LACPDU are
        transmitted. A value of '1' indicates Active mode where
        LACPDU are sent regardless of partner's lacp-activity
        value. A value of '0' indicates Passive mode where
        LACPDU are sent only when the partner's lacp-activity
        value is '1' (partner is in Active mode).";
    }
    bit lacp-timeout {
      position 2;
    }
  }
}
```

```
    description
      "Provides administrative control over the frequency of
      received LACPDUs. A value of '1' indicates Short Timeout
      (so partner uses frequent transmission). A value of '0'
      indicates Long Timeout (so partner can use infrequent
      transmission).";
  }
  bit aggregation {
    position 3;
    description
      "Provides administrative control over whether this
      Aggregation Port can be in a LAG with more than one
      member. A value of '1' indicates the port can be
      aggregated with other ports. A value of '0' indicates
      the port can only be a solitary link.";
  }
  bit synchronization {
    position 4;
    description
      "The Synchronization state of the MUX state machine.";
  }
  bit collecting {
    position 5;
    description
      "The Collecting state of the MUX state machine.";
  }
  bit distributing {
    position 6;
    description
      "The Distributing state of the MUX state machine.";
  }
  bit defaulted {
    position 7;
    description
      "Indicates the port is using the partner-admin values
      to select an Aggregator.";
  }
  bit expired {
    position 8;
    description
      "The Expired state of the Receive state machine.";
  }
}
description
  "LACP state values as transmitted in LACPDUs.
  Administrative control over the values of lacp-activity,
  lacp-timeout, aggregation, and (in partner-admin-lacp-state)
  synchronization is allowed.
  The remaining bits are read-only.";
reference
  "6.4.1, 6.4.2.3 of IEEE Std 802.1AX";
}

typedef bridge-reserved-addresses {
  type enumeration {
    enum nearest-customer-bridge {
      description
        "Reserved MAC address 01-C2-80-00-00";
    }
    enum slow-protocols-multicast {
      description
        "Reserved MAC address 01-C2-80-00-02";
    }
    enum nearest-non-tpmr-bridge {
      description
        "Reserved MAC address 01-C2-80-00-03";
    }
    enum nearest-bridge {
      description
        "Reserved MAC address 01-C2-80-00-0E";
    }
  }
}
```

```
description
  "Group addresses that are filtered by some or all types of
  bridges in order to limit the scope of propagation of PDUs
  in a Bridged Network.";
reference
  "8.6.3 of IEEE Std 802.1Q";
}

grouping link-map {
  description
    "Specifies the contents of the admin-conv-link-map.";
  reference
    "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
  choice method {
    default "cid-list";
    description
      "Provides two ways to specify the map contents.";
    leaf pattern {
      type identityref {
        base link-map-patterns;
      }
      description
        "Use a predefined pattern to fill the map.";
    }
    list cid-list {
      key "cid";
      description
        "Data structure to map a Conversation Identifier
        (CID) to a Link Number. Each entry consists of a CID
        and a list of link numbers that can potentially be
        selected for that CID. An empty list of link-numbers
        means that no links are selected for the CID.";
      leaf cid {
        type uint16 {
          range "0..4095";
        }
        description
          "Port Conversation Identifier";
      }
      leaf-list link-numbers {
        type uint16;
        description
          "List of zero or more Link Numbers that can
          potentially be selected for distribution of frames
          with this CID.";
      }
    }
  }
}

grouping service-map {
  description
    "Specifies the contents of the admin-conv-service-map.";
  reference
    "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
  choice method {
    default "cid-list";
    description
      "Provides two ways to specify the map contents.";
    leaf pattern {
      type identityref {
        base service-map-patterns;
      }
      description
        "Use a predefined pattern to fill the map.";
    }
    list cid-list {
      key "cid";
      description
        "Data structure to map service identifiers to
        conversation identifiers. Each entry consists of a
        Conversation ID (CID) and a list of zero or more
```

```
        Service Identifiers (SIDs) that map to it. An empty
        list of SIDs means there are no SIDs that map to this
        CID, and results in the same behavior as not having an
        entry for this CID.";
    leaf cid {
        type uint16 {
            range "0..4095";
        }
        description
            "Port Conversation Identifier";
    }
    leaf-list service-ids {
        type uint32;
        description
            "List of zero or more SIDs that map to the CID.";
    }
}
}
```

10.7.2 The *ieee802-dot1ax-linkagg* YANG module

```
module ieee802-dot1ax-linkagg {
    yang-version 1.1;
    namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-linkagg";
    prefix dot1ax;

    import ieee802-dot1ax-types {
        prefix ax;
    }
    import ieee802-types {
        prefix ieee;
    }
    import ietf-yang-types {
        prefix yang;
    }
    import ietf-interfaces {
        prefix if;
    }
    import iana-if-type {
        prefix ianaif;
    }

    organization
        "IEEE 802.1 Working Group";
    contact
        "WG-URL: http://www.ieee802.org/1/
        WG-Email: stds-802-1-l@ieee.org

        Contact: IEEE 802.1 Working Group Chair
        Postal: C/O IEEE 802.1 Working Group
                IEEE Standards Association
                45 Hoes Lane
                Piscataway, NJ 08854
                USA
        E-mail: stds-802-1-chairs@ieee.org";
    description
        "This YANG module describes the configuration model for Link
        Aggregation, as specified in IEEE Std 802.1AX, including Link
        Aggregation Control Protocol (LACP) and Conversation Sensitive
        Collection and Distribution.

        Copyright (C) IEEE (2025).

        This version of this YANG module is part of IEEE Std 802.1AX;
        see the standard itself for full legal notices.";

    revision 2025-03-11 {
```

```
description
  "Published as part of IEEE Std 802.1AXdz-202y.
  The following reference statement identifies each referenced IEEE
  Standard as updated by applicable amendments.";
reference
  "IEEE Std 802.1AX-2020 Link Aggregation:
  IEEE Std 802.1AXdz-202y.";
}

typedef key-group-ref {
  type leafref {
    path "/dotlax:linkagg/dotlax:key-group/dotlax:name";
  }
  description
    "This type is used by aggregators and aggregation ports to
    reference an entry in the key-group list.";
}

typedef agg-system-ref {
  type leafref {
    path "/dotlax:linkagg/dotlax:agg-system/dotlax:name";
  }
  description
    "This type is used in the key-group list entries to
    reference an entry in the agg-system list.";
}

augment "/if:interfaces/if:interface" {
  when "derived-from-or-self(if:type,'ianaif:ieee8023adLag')" {
    description
      "Applies to interfaces representing a LAG.";
  }
  description
    "Augment Interface with Aggregator parameters.";
  container lag {
    description
      "Contains the configuration and status information that
      allows an instance of an Aggregator to be managed.";
    leaf key-group-name {
      type key-group-ref;
      mandatory true;
      description
        "Specifies the entry in the link-aggregation key-group
        list to which this aggregator is assigned.";
    }
    leaf mac-address {
      type ieee:mac-address;
      config false;
      description
        "The MAC address assigned to the Aggregator.";
      reference
        "7.3.1.1.9 of IEEE Std 802.1AX";
    }
    container lacp {
      config false;
      description
        "Contains aggregator LACP operational data.";
      leaf actor-oper-key {
        type uint16;
        description
          "The current operational value of the Key for the
          Aggregator. The meaning of particular Key
          values is of local significance.";
        reference
          "7.3.1.1.8 of IEEE Std 802.1AX";
      }
    }
    leaf partner-system {
      type ieee:mac-address;
      description
        "The value of the MAC address portion of the
        System Identifier for the current
        protocol partner of this Aggregator. A value
```

```
        of zero indicates that there is no known partner.
        If the aggregation is manually configured, this
        System identifier value is assigned by the
        local System.";
    reference
        "7.3.1.1.10 of IEEE Std 802.1AX";
}
leaf partner-system-priority {
    type uint16;
    description
        "Indicates the priority value portion of the current
        Partner's System Identifier. If the aggregation is
        manually configured, this System Priority value is
        assigned by the local System.";
    reference
        "7.3.1.1.11 of IEEE Std 802.1AX";
}
leaf partner-oper-key {
    type uint16;
    description
        "The current operational value of the Key for the
        Aggregators current protocol Partner. If the
        aggregation is manually configured, this Key value
        is assigned by the local System.";
    reference
        "7.3.1.1.12 of IEEE Std 802.1AX";
}
leaf aggregate-or-individual {
    type boolean;
    description
        "Indicates whether the Aggregator represents an
        Aggregate (TRUE) or an Individual link (FALSE).";
    reference
        "7.3.1.1.6 of IEEE Std 802.1AX";
}
}
container cscd {
    if-feature "ax:cscd";
    config false;
    description
        "Aggregator parameters obtained by the operation of LACP
        supporting CSCD.";
    leaf partner-port-algorithm {
        type identityref {
            base ax:distribution-algorithm;
        }
        description
            "Operational value of the distribution algorithm in
            use by the LACP Partner.";
        reference
            "7.3.1.1.40 of IEEE Std 802.1AX";
    }
    leaf partner-conv-service-digest {
        type binary;
        description
            "The MD5 Digest of the admin-conv-service-map in use
            by the LACP Partner.";
        reference
            "7.3.1.1.42, 6.6.3.1 of IEEE Std 802.1AX";
    }
    leaf partner-conv-link-digest {
        type binary;
        description
            "The MD5 Digest of the admin-conv-link-map in use
            by the LACP Partner.";
        reference
            "7.3.1.1.41, 6.6.3.1 of IEEE Std 802.1AX";
    }
}
leaf discard-wrong-conversation {
    type boolean;
    description
        "The operational value that determines whether an
```

```

        description
            "Aggregator discards a frame that is collected
            from an Aggregation Port that is different from the
            Aggregation Port to which the Aggregator would
            distribute a frame with the same Port Conversation
            ID.";
        reference
            "7.3.1.1.37, 6.6 of IEEE Std 802.1AX";
    }
    leaf-list active-lag-links {
        type uint16;
        description
            "A list, possibly empty, of the operational
            link-number of each Aggregation Port active
            (i.e. Collecting) on this Aggregator.";
        reference
            "7.3.1.1.43 of IEEE Std 802.1AX";
    }
}
}
}

augment "/if:interfaces/if:interface/if:statistics" {
    when '../dot1ax:lag' {
        description
            "Applies to aggregators.";
    }
    description
        "Augment interface statistics with aggregator statistics.";
    container lag-stats {
        config false;
        description
            "Contains the set of stats associated with the
            Aggregator.";
        leaf unknown-protocol-frames {
            type yang:counter64;
            description
                "A count of data frames discarded on reception by all
                ports that are (or have been) members of the
                aggregation, due to the detection of an unknown Slow
                Protocols PDU (7.3.3.1.5)";
            reference
                "7.3.1.1.29 of IEEE Std 802.1AX";
        }
    }
}

augment "/if:interfaces/if:interface" {
    description
        "Augment interface model with Aggregation port
        configuration nodes.";
    container aggport {
        presence
            "When present, this interface supports Link Aggregation";
        description
            "Contains Aggregation Port configuration related nodes,
            which provides the basic management controls necessary
            to allow an instance of an Aggregation Port to be managed,
            for the purposes of Link Aggregation.";
        leaf key-group-name {
            type key-group-ref;
            mandatory true;
            description
                "Specifies the entry in the link-aggregation key-group
                list to which this aggregation-port is assigned.";
        }
        leaf actor-port-number {
            type uint16 {
                range "1..65535";
            }
            description
                "The port number assigned to the Aggregation Port.
                The port number is communicated in LACPDUs as the

```



```
    Actor Port.
    This leaf provides the ability to administratively
    override the initial value provided by the system.";
  reference
    "7.3.2.1.14, 6.4.6 of IEEE Std 802.1AX";
}
leaf actor-port-priority {
  type uint16;
  default "0x8000";
  description
    "The priority value assigned to this Aggregation Port.";
  reference
    "7.3.2.1.15, 6.4.6 of IEEE Std 802.1AX";
}
leaf partner-admin-key {
  type uint16 {
    range "1..65535";
  }
  description
    "The current administrative value of the Key for the
    protocol Partner. The assigned value is used, along
    with the value of port-partner-admin-system-priority,
    partner-admin-system, partner-admin-port, and
    partner-admin-port-priority, in order to achieve
    manually configured aggregation.
    The default value is the initial value of actor-port-
    number provided by the system.";
  reference
    "7.3.2.1.10 of IEEE Std 802.1AX";
}
leaf partner-admin-port {
  type uint16 {
    range "1..65535";
  }
  description
    "The current administrative value of the port number for
    the protocol Partner. The assigned value is used, along
    with the value of partner-admin-system-priority,
    partner-admin-system, port-partner-admin-key, and
    partner-admin-port-priority, in order to achieve
    manually configured aggregation.
    The default value is the initial value of actor-port-
    number provided by the system.";
  reference
    "7.3.2.1.16 of IEEE Std 802.1AX";
}
leaf partner-admin-port-priority {
  type uint16;
  default "0";
  description
    "The current administrative value of the port priority
    for the protocol Partner. The assigned value is used,
    along with the value of partner-admin-system-priority,
    partner-admin-system, partner-admin-key, and
    partner-admin-port, in order to achieve manually
    configured aggregation.";
  reference
    "7.3.2.1.18 of IEEE Std 802.1AX";
}
leaf partner-admin-state {
  type ax:lacp-state {
    bit lacp-activity {
      position 1;
      description
        "When the LACP_Activity state is set to '0', the
        transmission of LACPDUs is controlled by the
        actor-admin-state.LACP_Activity.";
    }
    bit lacp-timeout {
      position 2;
      description
        "When the LACP_Timeout is set to '0', LACPDUs
```

```
        are transmitted at the Slow_Periodic_Time.";
    }
    bit aggregation {
        position 3;
        description
            "When the Aggregation state is set to '0', this
            port cannot be aggregated with any other port.";
    }
    bit synchronization {
        position 4;
        description
            "When the Synchronization state is set to '0', this
            port cannot become active.";
    }
}
default "synchronization";
description
    "Provides administrative control over the partner's
    LACP_Activity, LACP_Timeout, Aggregation, and
    Synchronization state when the partner's information is
    unknown (i.e. no LACPDUs are received from the partner).";
reference
    "7.3.2.1.22, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
}
leaf wtr-time {
    type uint16 {
        range "0..32767";
    }
    default "0";
    description
        "The wait-to-restore (WTR) period, in seconds, that
        needs to elapse between an Aggregation Port on a LAG
        coming up (Port_Operational becoming TRUE) and being
        permitted to become active (transmitting and
        receiving frames) on the LAG.";
    reference
        "7.3.2.1.30 of IEEE Std 802.1AX";
}
leaf wtr-revertive {
    type boolean;
    default "true";
    description
        "Controls revertive or non-revertive mode of operation.
        When TRUE, the Aggregation Port can become active as
        soon as the wait-to-restore timer expires regardless of
        the state of other links in the LAG.
        When FALSE, the Aggregation Port cannot become active
        unless there are no other links that can become active
        in the LAG. The default value is TRUE.";
    reference
        "7.3.2.1.31 of IEEE Std 802.1AX";
}
container lacp {
    config false;
    description
        "Contains Aggregation port LACP operational related
        nodes.";
    leaf actor-lacp-version {
        type uint8;
        description
            "The version number transmitted in LACPDUs on this
            Aggregation Port";
        reference
            "7.3.2.1.33 of IEEE Std 802.1AX";
    }
    leaf actor-oper-key {
        type uint16;
        description
            "The current operational value of the Key for the
            Aggregation Port. The meaning of particular Key values
            is of local significance.";
        reference

```

```
    "7.3.2.1.5 of IEEE Std 802.1AX";
}
leaf actor-oper-state {
  type ax:lacp-state;
  description
    "The operational value of the Actor_State as
    transmitted in LACPDU.";
  reference
    "7.3.2.1.21, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
}
leaf partner-lacp-version {
  type uint8;
  description
    "The version number in the LACPDU most recently
    received on this Aggregation Port.";
  reference
    "7.3.2.1.34 of IEEE Std 802.1AX";
}
leaf partner-oper-system-priority {
  type uint16;
  description
    "Indicates the operational value of priority associated
    with the Partners System ID. The value of this
    attribute can contain the manually configured value
    carried in partner-admin-system-priority if there is
    no protocol Partner.";
  reference
    "7.3.2.1.7 of IEEE Std 802.1AX";
}
leaf partner-oper-system {
  type ieee:mac-address;
  description
    "Represents the MAC address portion of the Aggregation
    Port's current protocol partner's System Identifier.
    A value of zero indicates that there is no known
    protocol Partner.
    The value of this attribute can contain the manually
    configured value carried in partner-admin-system if
    there is no protocol Partner.";
  reference
    "7.3.2.1.9 of IEEE Std 802.1AX";
}
leaf partner-oper-key {
  type uint16;
  description
    "The current operational value of the Key for the
    protocol Partner. The value of this attribute can
    contain the manually configured value carried in
    partner-admin-key if there is no protocol Partner.";
  reference
    "7.3.2.1.11 of IEEE Std 802.1AX";
}
leaf partner-oper-port {
  type uint16;
  description
    "The operational port number assigned by the
    Aggregation Port's protocol Partner. The value of this
    attribute can contain the administratively configured
    value carried in partner-admin-port if there is no
    protocol Partner.";
  reference
    "7.3.2.1.17 of IEEE Std 802.1AX";
}
leaf partner-oper-port-priority {
  type uint16;
  description
    "The operational priority value assigned by the
    Aggregation Port's protocol Partner. The value of this
    attribute can contain the administratively configured
    value carried in partner-admin-port-priority if there
    is no protocol Partner.";
  reference
```

```
    "7.3.2.1.19 of IEEE Std 802.1AX";
  }
  leaf partner-oper-state {
    type ax:lacp-state;
    description
      "The operational value of the partner's LACP state
      derived from received LACPDUs or, when Defaulted is
      true, from the partner-admin-state.";
    reference
      "7.3.2.1.23, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
  }
  leaf aggregate-or-individual {
    type boolean;
    description
      "When true indicates the Aggregation Port can join a
      LAG consisting of multiple Aggregation Ports.
      When false, indicates that the Aggregation Port can
      only operate as an Solitary link because the
      Aggregation bit is false in either
      actor-oper-port-state or partner-oper-port-state.";
    reference
      "7.3.2.1.24 of IEEE Std 802.1AX";
  }
  leaf wtr-waiting {
    type boolean;
    description
      "Indicates the Aggregation Port is inhibited from
      becoming active for an interval (determined by
      wtr-time) after becoming operational or while
      non-revertive operation is being enforced by the
      Selection Logic.";
    reference
      "7.3.2.32 of IEEE Std 802.1AX";
  }
}
container cscd {
  if-feature "ax:cscd";
  description
    "Aggregation port parameters for support of CSCD.";
  leaf admin-link-number {
    type uint16;
    description
      "The Link Number value for the Aggregation Port,
      that is unique among all Aggregation Ports in the same
      key group. This leaf provides the ability to
      administratively override the initial value provided
      by the system. A value of 0 is allowed, but
      results in no frames being distributed to this
      Aggregation Port.
      The default value is the initial value of actor-port-
      number provided by the system.";
    reference
      "7.3.2.1.27, 6.6.3.2 of IEEE Std 802.1AX";
  }
  leaf partner-link-number {
    type uint16;
    config false;
    description
      "The last received value of the Partner Link Number,
      or zero if the Aggregation Port is using default
      values for the Partner or the Partner LACP Version
      is 1.";
    reference
      "7.3.2.1.29 of IEEE Std 802.1AX";
  }
  leaf link-number {
    type uint16;
    config false;
    description
      "The operational link number for this Aggregation Port.
      The value is either the same as the admin-link-number,
      or the corresponding value of the LACP partner.";
```

```
        reference
        "7.3.2.1.28 of IEEE Std 802.1AX";
    }
}
}
}

augment "/if:interfaces/if:interface/if:statistics" {
    when '../dotlax:aggport' {
        description
        "Applies to aggregation ports.";
    }
    description
    "Augment interface statistics with aggport statistics.";
    container aggport-stats {
        config false;
        description
        "Contains stats associated with the Aggregation Port.";
        leaf lacp-pdu-rx {
            type yang:counter64;
            description
            "The number of valid LACPDUs received on this
            Aggregation Port.";
            reference
            "7.3.3.1.2 of IEEE Std 802.1AX";
        }
        leaf marker-pdu-rx {
            type yang:counter64;
            description
            "The number of valid Marker PDUs received on this
            Aggregation Port.";
            reference
            "7.3.3.1.3 of IEEE Std 802.1AX";
        }
        leaf marker-response-pdu-rx {
            type yang:counter64;
            description
            "The number of valid Marker Response PDUs received on
            this Aggregation Port.";
            reference
            "7.3.3.1.4 of IEEE Std 802.1AX";
        }
        leaf unknown-rx {
            type yang:counter64;
            description
            "The number of frames received that either:
            a) Carry the Slow Protocols Ethernet Type value,
            but contain an unknown PDU, or
            b) Are addressed to the Slow Protocols group MAC
            Address, but do not carry the Slow Protocols
            Ethernet Type.";
            reference
            "7.3.3.1.5 of IEEE Std 802.1AX";
        }
        leaf illegal-rx {
            type yang:counter64;
            description
            "The number of frames received that carry the Slow
            Protocols Ethernet Type value, but contain a badly
            formed PDU or an illegal value of Protocol Subtype.";
            reference
            "7.3.3.1.6 of IEEE Std 802.1AX";
        }
        leaf lacp-pdu-tx {
            type yang:counter64;
            description
            "The number of LACPDUs transmitted on this
            Aggregation Port.";
            reference
            "7.3.3.1.7 of IEEE Std 802.1AX";
        }
        leaf marker-pdu-tx {
```

```
    type yang:counter64;
    description
      "The number of Marker PDUs transmitted on this
      Aggregation Port.";
    reference
      "7.3.3.1.8 of IEEE Std 802.1AX";
  }
  leaf marker-response-pdu-tx {
    type yang:counter64;
    description
      "The number of Marker Response PDUs transmitted on
      this Aggregation Port.";
    reference
      "7.3.3.1.9 of IEEE Std 802.1AX";
  }
}

container linkagg {
  description
    "Link Aaggregation System specific configuration nodes.";
  list agg-system {
    key "name";
    description
      "List of aggregation systems.";
    leaf name {
      type string;
      description
        "Name for the aggregation system.";
    }
    leaf actor-system {
      type ieee:mac-address;
      description
        "The part of the System Identifier that is a globally
        unique MAC address. This leaf provides the ability to
        administratively override the initial value provided
        by the system.";
      reference
        "7.3.1.1.4, 7.3.2.1.3 of IEEE Std 802.1AX";
    }
    leaf actor-system-priority {
      type uint16;
      default "0x8000";
      description
        "The part of the System Identifier that is the
        priority of the system.";
      reference
        "7.3.1.1.5, 7.3.2.1.2 of IEEE Std 802.1AX";
    }
  }
  list key-group {
    key "name";
    unique "actor-admin-key";
    description
      "List of key groups. A key group is the set of aggregators
      and aggregation ports that share the same system priority,
      system identifier, and aggregation key, and therefore can
      potentially form a Link Aggregation Group. Each entry in
      the key group list contains the parameters common to all
      aggregation ports and/or aggregators in the key group.";
    reference
      "6.3.5, 6.4.12 of IEEE Std 802.1AX";
    leaf name {
      type string;
      description
        "Name for the key group.";
    }
    leaf actor-admin-key {
      type uint16 {
        range "1..65535";
      }
      mandatory true;
    }
  }
}
```

```
description
  "The administrative value of the key used by the
  Aggregators and Aggregation Ports in this key-group.";
reference
  "7.3.1.1.7, 7.3.2.1.4 of IEEE Std 802.1AX";
}
leaf agg-system-name {
  type agg-system-ref;
  mandatory true;
  description
    "Specifies the aggregation system for this key
    group.";
}
leaf actor-protocol-da {
  type ax:bridge-reserved-addresses;
  must ". = 'nearest-customer-bridge' or
        . = 'slow-protocols-multicast' or
        . = 'nearest-non-tpmr-bridge'" {
    error-message "Invalid protocol address";
  }
  default "slow-protocols-multicast";
  description
    "A 6-octet read-write MAC Address value specifying the DA
    to be used when sending Link Aggregation Control and
    Marker PDUs. Valid addresses are the
    nearest-customer-bridge, slow-protocols-multicast,
    and nearest-non-tpmr-bridge group addresses. The
    default value is the slow-protocols-multicast address.";
  reference
    "7.3.2.2.1, 6.2.10.2 of IEEE Std 802.1AX";
}
leaf collector-max-delay {
  type uint16;
  description
    "Specifies the maximum delay, in tens of microseconds,
    between receiving a frame from an Aggregator Port, and
    either delivering the frame to the Aggregator Client or
    discarding the frame. A value of zero means the delay
    is less than the minimum increment (< 10us).
    This leaf provides the ability to administratively
    override the initial value provided by the system.";
  reference
    "7.3.1.1.32, 6.2.3.1.1, B.3 of IEEE Std 802.1AX";
}
leaf actor-admin-state {
  type ax:lacp-state {
    bit lacp-activity {
      position 1;
      description
        "Setting the LACP_Activity state to '0' means that the
        transmission of LACPDUs is controlled by the value of
        partner-oper-state.LACP_Activity.";
    }
    bit lacp-timeout {
      position 2;
      description
        "Setting the LACP_Timeout to '0' means that actor uses
        the Long_Timeout value, allowing the partner to transmit
        LACPDUs at the Slow_Periodic_Time.";
    }
    bit aggregation {
      position 3;
      description
        "Setting the Aggregation state to '0' prevents this
        port from being aggregated with any other ports.";
    }
  }
  default "lacp-activity lacp-timeout aggregation";
  description
    "Provides administrative control over the values of the
    LACP_Activity, LACP_Timeout, and Aggregation state.";
  reference
```

```
    "7.3.2.1.20, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
}
leaf partner-admin-system {
    type ieee:mac-address;
    default "00-00-00-00-00-00";
    description
        "The administrative value of the MAC address portion of
        the Partner's System Identifier.
        The assigned value is used, along with the value of
        port-partner-admin-system, partner-admin-key,
        partner-admin-port, and partner-admin-port-priority,
        to achieve administratively configured Link
        Aggregation Groups with a partner that does not run
        LACP.";
    reference
        "7.3.2.1.8 of IEEE Std 802.1AX";
}
leaf partner-admin-system-priority {
    type uint16;
    default "0";
    description
        "The administrative value of priority portion of the
        the Partner's System Identifier. The assigned
        value is used, along with the value of
        port-partner-admin-system, partner-admin-key,
        partner-admin-port, and partner-admin-port-priority,
        to achieve administratively configured Link
        Aggregation Groups with a partner that does not run
        LACP.";
    reference
        "7.3.2.1.6 of IEEE Std 802.1AX";
}
leaf port-algorithm {
    type identityref {
        base ax:distribution-algorithm;
    }
    default "ax:unspecified";
    description
        "Identifies the algorithm used by the Aggregator to
        assign frames to a Port Conversation ID. Default is
        the value for an unspecified distribution algorithm.
        When the identity description specifies a 4 octet value,
        this value will be used for cscd purposes, and included
        in LACPDUs. Otherwise this leaf is used purely for
        local selection of a distribution algorithm, and the
        unspecified distribution algorithm is used for cscd
        purposes.";
    reference
        "7.3.1.1.33, Table 8-1 of IEEE Std 802.1AX";
}
leaf-list lags {
    type if:interface-ref;
    config false;
    description
        "A list of the if:name of aggregators assigned to this
        key group.";
    reference
        "linkagg:key-group";
}
leaf-list aggports {
    type if:interface-ref;
    config false;
    description
        "A list of the if:name of aggregation ports assigned to
        this key group.";
    reference
        "linkagg:key-group";
}
container cscd {
    if-feature "ax:cscd";
    description
        "Contains CSCD parameters that need to be consistent for
```



```
    all aggregation ports and aggregators in the key group.";
  container admin-conv-service-map {
    if-feature "ax:sid-map";
    description
      "Data structure to map service identifiers to
      conversation identifiers. Each entry consists of a
      Conversation ID (CID) and a list of zero or more
      Service Identifiers (SIDs) that map to it. An empty
      list of SIDs means there are no SIDs that map to this
      CID, and results in the same behavior as not having an
      entry for this CID.";
    reference
      "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
    uses ax:service-map;
  }
  leaf admin-conv-service-digest {
    type binary;
    config false;
    description
      "The MD5 Digest of the admin-conv-service-map. The
      value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
      when the distribution algorithm specified by
      agg-port-algorithm does not use the
      admin-conv-service-map.";
    reference
      "7.3.1.1.39, 6.6.3.1 of IEEE Std 802.1AX";
  }
  container admin-conv-link-map {
    description
      "Data structure to map a Conversation Identifier
      (CID) to a Link Number. Each entry consists of a CID
      and a list of link numbers that can potentially be
      selected for that CID. An empty list of link-numbers
      means that no links are selected for the CID.";
    reference
      "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
    uses ax:link-map;
  }
  leaf admin-conv-link-digest {
    type binary;
    config false;
    description
      "The MD5 Digest of the admin-conv-link-map. The value
      value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
      when the distribution algorithm specified by
      agg-port-algorithm does not use the
      admin-conv-link-map.";
    reference
      "7.3.1.1.38, 6.6.3.1 of IEEE Std 802.1AX";
  }
  leaf admin-discard-wrong-conv {
    if-feature "ax:dwc";
    type enumeration {
      enum force-true {
        value 1;
        description
          "Indicates that an Aggregator discards a
          frame that is collected from an Aggregation Port
          that is different from the Aggregation Port to
          which the Aggregator would distribute a frame
          with the same Port Conversation ID.";
      }
      enum force-false {
        value 2;
        description
          "Indicates that an Aggregator does not discard
          a frame that is collected from an Aggregation Port
          that is different from the Aggregation Port to
          which the Aggregator would distribute a frame with
          the same Port Conversation ID. This is the behavior
          of the Aggregator when DWC is not supported";
      }
    }
  }
}
```

```
enum auto {  
    value 3;  
    description  
        "Indicates that the Aggregator behaves as  
        if the value was force-true only when the actor  
        and partner agree on the algorithms (other than  
        unspecified) and mapping tables used to map frames  
        to Aggregation Ports, and behaves as if the value  
        was force-false otherwise.";  
}  
}  
default "force-false";  
description  
    "Indicates whether an Aggregator discards a  
    frame that is collected from an Aggregation Port  
    that is different from the Aggregation Port to which  
    the Aggregator would distribute a frame with the  
    same Port Conversation ID.";  
reference  
    "7.3.1.1.35, 6.6 of IEEE Std 802.1AX";  
}  
}  
}  
}
```

10.7.3 The *ieee802-dot1ax-drni* YANG module

```
module ieee802-dot1ax-drni {  
    yang-version 1.1;  
    namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-drni";  
    prefix dot1ax-drni;  
  
    import ieee802-dot1ax-types {  
        prefix ax;  
    }  
    import ieee802-dot1ax-linkagg {  
        prefix dot1ax;  
    }  
    import ieee802-types {  
        prefix ieee;  
    }  
    import ietf-yang-types {  
        prefix yang;  
    }  
    import ietf-interfaces {  
        prefix if;  
    }  
  
    organization  
        "IEEE 802.1 Working Group";  
    contact  
        "WG-URL: http://www.ieee802.org/1/  
        WG-Email: stds-802-1-1@ieee.org  
  
        Contact: IEEE 802.1 Working Group Chair  
        Postal: C/O IEEE 802.1 Working Group  
        IEEE Standards Association  
        45 Hoes Lane  
        Piscataway, NJ 08854  
        USA  
        E-mail: stds-802-1-chairs@ieee.org";  
    description  
        "This YANG module describes the configuration model for a  
        Distributed Resilient Network Interface (DRNI) as specified  
        in 802.1AX.  
  
        Copyright (C) IEEE (2025)."
```

This version of this YANG module is part of IEEE Std 802.1AX;
see the standard itself for full legal notices.";

```
revision 2025-03-11 {
  description
    "Published as part of IEEE Std 802.1AXdz-202y.
    The following reference statement identifies each referenced IEEE
    Standard as updated by applicable amendments.";
  reference
    "IEEE Std 802.1AX-2020 Link Aggregation:
    IEEE Std 802.1AXdz-202y.";
}

typedef irp-state {
  type bits {
    bit reserved-1 {
      position 1;
      description
        "Reserved for future use. It is set to 0 on
        transmit and ignored on receipt.";
    }
    bit reserved-2 {
      position 2;
      description
        "Reserved for future use. It is set to 0 on
        transmit and ignored on receipt.";
    }
    bit short-timeout {
      position 3;
      description
        "The Short_Timeout flag indicates the Timeout control value
        in use by the DRCP Receive machine on this IRP. Short Timeout
        is encoded as a 1; Long Timeout is encoded as a 0.";
    }
    bit synchronization {
      position 4;
      description
        "When the Sync flag is TRUE (1), the DRCP Receive machine has
        determined the Neighbor DRNI System has a compatible
        configuration for forming a DRNI.";
    }
    bit irc-data {
      position 5;
      description
        "When the IRC Data flag is TRUE (1), the transfer of Up
        and Down frames is permitted on the IRC.";
    }
    bit drni {
      position 6;
      description
        "The DRNI flag is TRUE (1) when this DRNI System is paired
        with another DRNI System and FALSE (0) otherwise.";
    }
    bit defaulted {
      position 7;
      description
        "When the Defaulted flag is TRUE (1), the DRCP Receive machine
        is using default operational Neighbor information.
        When FALSE (0), the operational Neighbor information
        in use has been received in a DRCPDU.";
    }
    bit expired {
      position 8;
      description
        "When the Expired flag is TRUE (1), the DRCP Receive machine
        is in the EXPIRED state.";
    }
  }
  description
    "IRP state values as transmitted in DRCPDUs.
    Administrative control over the values of Short_Timeout
    and IRC_Data is allowed. The remaining bits are read-only.";
```

```

reference
  "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
}

grouping drni-mask {
  description
    "Specifies the contents of a bit mask indexed by CID.";
  leaf-list cid-str {
    type string {
      pattern '(0|[1-9][0-9]{0,3})'
        + '((-0|[1-9][0-9]{0,3})'
        + '(%0|[1-9][0-9]{0,3}))?)?'
        + '(',
      pattern '(0|[1-9][0-9]{0,3})'
        + '((-0|[1-9][0-9]{0,3})'
        + '(%0|[1-9][0-9]{0,3}))?)?'
        + ')*';
    }
    default "0-4095";
    description
      "A comma separated list of single CID values (x),
      CID ranges (x-y), or ranges with an increment (x-y%z).
      For example, the string '0,10-13,100-110%4' sets the
      bits in the mask corresponding to CIDs
      0,10,11,12,13,100,104,108.
      Some special cases are:
      'x-y%z' where x+z>y is the same as 'x'.
      'x-y%0' is the same as 'x'.
      'x-y%1' is the same as 'x-y'.
      'x-y%2' sets the bit for all even CIDs in the range
      when x is even, and for all odd CIDs when x is odd.
      Although the pattern allows values of x, y, and z in the
      range 0-9999, resulting CID values greater than 4095
      have no effect on the mask.
      Specified as a list of strings, so that long strings
      can be broken into several shorter strings.
      The default sets all bits in the mask.";
  }
  leaf invert-mask {
    type boolean;
    default "false";
    description
      "When true the specified mask is inverted:
      each zero replaced with a one,
      and each one replaced with a zero.";
  }
}

augment "/if:interfaces/if:interface/dot1ax:lag" {
  description
    "Augmentation parameters only for Aggregators with
    DRNI enabled.";
  container drni {
    presence "When present, this Aggregator is enabled for DRNI";
    description
      "Aggregator parameters to support a Distributed
      Resilient Network Interface";
    leaf irp-name {
      type if:interface-ref;
      mandatory true;
      description
        "Interface Name (if:name) of the Port supporting the
        Intra Relay Port (IRP) of this DRNI Gateway.";
      reference
        "7.4.1.1.23 of IEEE Std 802.1AX";
    }
    leaf drni-aggregator-key {
      type uint16 {
        range "1..65535";
      }
      description
        "The Aggregator Key value to be used by the Aggregator

```

```

        supporting this DRNI Gateway (and the Aggregation Ports
        assigned to this DRNI Gateway) when paired with a
        neighbor DRNI System via the IRC. The default value is
        the Aggregator's Actor_Admin_Aggregator_Key value.";
    reference
        "7.4.1.1.15 of IEEE Std 802.1AX";
}
leaf drni-aggregator-system {
    type ieee:mac-address;
    default "00-00-00-00-00-00";
    description
        "The Aggregator System value to be used by the
        Aggregator supporting this DRNI Gateway (and the
        Aggregation Ports assigned to this DRNI Gateway)
        when paired with a neighbor DRNI System via the
        Intra-Relay Connection (IRC).";
    reference
        "7.4.1.1.13 of IEEE Std 802.1AX";
}
leaf drni-aggregator-system-priority {
    type uint16;
    default "0";
    description
        "The Aggregator System Priority value to be used by the
        Aggregator supporting this DRNI Gateway (and the
        Aggregation Ports assigned to this DRNI Gateway) when
        paired with a neighbor DRNI System via the IRC.";
    reference
        "7.4.1.1.14 of IEEE Std 802.1AX";
}
leaf drcp-protocol-da {
    type ax:bridge-reserved-addresses;
    must ". = 'nearest-customer-bridge' or
        . = 'nearest-bridge' or
        . = 'nearest-non-tpmr-bridge'" {
        error-message "Invalid protocol address";
    }
    default "nearest-non-tpmr-bridge";
    description
        "A 6-octet read-write MAC Address value specifying the
        Destination Address for Distributed Relay Control PDUs
        transmitted on the Intra-Relay Port. Valid addresses are
        the nearest-customer-bridge, nearest-bridge, and
        nearest-non-tpmr-bridge group addresses. The default
        value is the nearest-non-tpmr-bridge group
        address.";
    reference
        "7.4.1.1.12, 9.6.1.1 of IEEE Std 802.1AX";
}
leaf home-admin-irp-state {
    type irp-state {
        bit short-timeout {
            position 3;
            description
                "The Short_Timeout flag indicates the Timeout control value
                in use by the DRCP Receive machine on this IRP. Short Timeout
                is encoded as a 1; Long Timeout is encoded as a 0.";
        }
        bit irc-data {
            position 5;
            description
                "When the IRC_Data flag is TRUE (1), the transfer of Up
                and Down frames is permitted on the IRC.";
        }
    }
    default "short-timeout irc-data";
    description
        "Provides administrative control over the values of the
        Short_Timeout and IRC-Data state.";
    reference
        "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
}

```

```
leaf home-oper-irp-state {
  type irp-state;
  config false;
  description
    "A string of 8 bits, corresponding to the current
    operational value of IRP_State as transmitted in
    DRCPDUs.";
  reference
    "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
}
leaf home-cscd-gateway-control {
  type boolean;
  default "true";
  description
    "When TRUE, allows the DRNI Gateway Port selection to
    be based on the CSCD parameters that control the
    Aggregator Port selection.";
  reference
    "7.4.1.1.16 of IEEE Std 802.1AX";
}
leaf home-dr-client-gateway-control {
  type boolean;
  default "true";
  description
    "When TRUE, allows the Distributed Relay Client to
    determine whether to forward frames through the DRNI
    Gateway Port.";
  reference
    "7.4.1.1.17 of IEEE Std 802.1AX";
}
leaf home-gateway-algorithm {
  type identityref {
    base ax:distribution-algorithm;
  }
  default "ax:unspecified";
  description
    "Identifies the algorithm used by the DRNI Gateway to
    assign frames to a Gateway Conversation ID. 8.2 provides
    the IEEE 802.1 OUI (00-80-C2) Gateway Algorithm
    encodings. Default is the value for an unspecified
    distribution algorithm.";
  reference
    "7.4.1.1.6 of IEEE Std 802.1AX";
}
container home-gateway-conv-service-map {
  if-feature "ax:sid-map";
  description
    "Data structure to map service identifiers to
    conversation identifiers. Each entry consists of a
    Conversation ID (CID) and a list of zero or more Service
    Identifiers (SIDs) that map to it. Frames with Service
    IDs not contained in the map are not mapped to any
    Gateway Conversation ID and are discarded.";
  reference
    "7.4.1.1.20, 6.6.3.1 of IEEE Std 802.1AX";
  uses ax:service-map;
}
leaf home-gateway-conv-service-digest {
  type binary;
  config false;
  description
    "The MD5 Digest of the home-gateway-conv-service-map. The
    value is NULL (AAAAAAAAAAAAAAAAAAAAA== in base64) when
    the distribution algorithm specified
    by agg-port-algorithm does not use the
    home-gateway-conv-service-map.";
  reference
    "7.4.1.1.21 of IEEE Std 802.1AX";
}
container gateway-enable-mask {
  description
    "A vector of Boolean values, with one value for each
```

```
    possible Gateway Conversation ID. A 1 indicates that
    frames associated with that Gateway Conversation ID
    are allowed to pass through this Gateway Port, and a
    0 indicates that such frames are not allowed to pass.
    Default value is all bits set to 1.";
  reference
    "7.4.1.1.18, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
  uses drni-mask;
}
container gateway-preference-mask {
  description
    "A vector of Boolean values, with one value for each
    possible Gateway Conversation ID. A 1 indicates that
    this Gateway Port is the preferred Gateway when both
    DRNI Gateways have the Gateway Conversation ID enabled
    in the gateway-available-mask, and a 0 indicates that
    it is not preferred.
    Default value is all bits set to 1.";
  reference
    "7.4.1.1.19, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
  uses drni-mask;
}
container gateway-available-mask {
  config false;
  description
    "A vector of Boolean values, with one value for each
    possible Gateway Conversation ID. A 1 indicates that
    this Gateway Port is eligible to be selected to pass
    that Gateway Conversation ID, and a 0 indicates that
    it is not eligible.";
  reference
    "7.4.1.1.22, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
  uses drni-mask;
}
container neighbor {
  config false;
  description
    "Operational values for the DRNI neighbor obtained
    from DRCPDUs.";
  leaf oper-irp-state {
    type irp-state;
    description
      "A string of 8 bits, corresponding to the current
      operational value of IRP_State as transmitted in
      DRCPDUs.";
    reference
      "7.4.1.1.26, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
  }
  leaf system {
    type ieee:mac-address;
    description
      "The MAC Address portion of the System Identifier of
      the Neighbor DRNI System (connected via the
      Intra-Relay Port). ";
    reference
      "7.4.1.1.29 of IEEE Std 802.1AX";
  }
  leaf system-priority {
    type uint16;
    description
      "The priority portion of the System Identifier of the
      Neighbor DRNI System (connected via the Intra-Relay
      Port).";
    reference
      "7.4.1.1.30 of IEEE Std 802.1AX";
  }
  leaf drni-key {
    type uint16;
    description
      "The DRNI key value received from the Neighbor DRNI
      System (connected via the IntraRelay Port).";
    reference
```

```
    "7.4.1.1.31 of IEEE Std 802.1AX";
  }
  leaf aggregator-algorithm {
    type identityref {
      base ax:distribution-algorithm;
    }
    description
      "The Port algorithm used by the Neighbor Aggregator to
       assign frames to Port Conversation IDs.";
    reference
      "7.4.1.1.33 of IEEE Std 802.1AX";
  }
  leaf aggregator-conv-service-digest {
    type binary;
    description
      "The MD5 Digest of the Neighbor Aggregator's
       Admin_Conv_Service_Map. Obtained from the Home
       Aggregator State TLV last received from the Neighbor
       DRNI System.";
    reference
      "7.4.1.1.34 of IEEE Std 802.1AX";
  }
  leaf aggregator-conv-link-digest {
    type binary;
    description
      "The MD5 Digest of the Neighbor Aggregator's
       Admin_Conv_Link_Map. Obtained from the Home Aggregator
       State TLV (9.6.2.4) last received from the Neighbor
       DRNI System.";
    reference
      "7.4.1.1.35 of IEEE Std 802.1AX";
  }
  leaf partner-system-priority {
    type uint16;
    description
      "The priority portion of the System Identifier of the
       Neighbor Aggregator's Partner.";
    reference
      "7.4.1.1.36 of IEEE Std 802.1AX";
  }
  leaf partner-system {
    type ieee:mac-address;
    description
      "The MAC Address portion of the System Identifier of
       the Neighbor Aggregator's Partner.";
    reference
      "7.4.1.1.37 of IEEE Std 802.1AX";
  }
  leaf partner-aggregator-key {
    type uint16;
    description
      "The operational key value of the Neighbor
       Aggregator's Partner.";
    reference
      "7.4.1.1.38 of IEEE Std 802.1AX";
  }
  leaf cscd-state {
    type bits {
      bit reserved-1 {
        position 0;
        description
          "Bit 1 is reserved for future use. It is set to 0
           and ignored on receipt.";
      }
      bit reserved-2 {
        position 1;
        description
          "Bit 2 is reserved for future use. It is set to 0
           and ignored on receipt.";
      }
      bit reserved-3 {
        position 2;
      }
    }
  }
```



```
        description
        "Bit 3 is reserved for future use. It is set to 0
        and ignored on receipt.";
    }
    bit cscd_gateway_control {
        position 3;
        description
        "CSCD_Gateway_Control is encoded in bit 4. When
        this flag is TRUE, the DRNI Gateway is configured
        to minimize forwarding data frames on the IRC by
        selecting the DRNI Gateway and Aggregator Ports
        for forwarding any given Conversation ID to be in
        the same DRNI System.";
    }
    bit discard_wrong_conversation {
        position 4;
        description
        "Discard_Wrong_Conversation is encoded in bit 5.
        The Aggregator's Discard_Wrong_Conversation
        value.";
    }
    bit differ_conv_link_digests {
        position 5;
        description
        "Differ_Conv_Link_Digests is encoded in bit 6.
        This flag is TRUE when the Aggregator's
        Actor_Conv_Link_Digest matches the Aggregator's
        Partner_Conv_Link_Digest.";
    }
    bit differ_conv_service_digests {
        position 6;
        description
        "Differ_Conv_Service_Digests is encoded in bit 7.
        This flag is TRUE when the Aggregator's
        Actor_Conv_Service_Digest matches the Aggregator's
        Partner_Conv_Service_Digest.";
    }
    bit differ_port_algorithms {
        position 7;
        description
        "Differ_Port_Algorithms is encoded in bit 8. The
        Aggregator's differPortAlgorithms flag is TRUE
        when the Aggregator's Actor_Port_Algorithm matches
        the Aggregator's Partner_Port_Algorithm.";
    }
}
description
"8 bits, corresponding to the Aggregator_CSCD_State
in the Neighbor_Aggregator_State variable. The first
three bits (the least significant bits of CSCD_State)
are reserved; the fourth bit corresponds to the
Neighbor's value for Home_Admin_CSCD_Gateway_Control;
the fifth bit corresponds to the Neighbor Aggregator's
operational value for Discard_Wrong_Conversation; and
the sixth, seventh, and eighth bits correspond to the
Neighbor Aggregator's operational value for
differConvLinkDigests, differConvServiceDigests, and
differPortAlgorithms, respectively, (the most
significant bits of CSCD_State).";
reference
"7.4.1.1.39 of IEEE Std 802.1AX";
}
leaf-list active-links {
    type uint16;
    description
    "A list of the operational Link_Numbers of Aggregation
    Ports that are currently active (i.e., collecting) on
    the Neighbor's Aggregator. An empty list indicates that
    there are no Aggregation Ports active. Each integer
    value in the list carries an aAggPortOperLinkNumber
    attribute value.";
    reference
```

```

        "7.4.1.1.40 of IEEE Std 802.1AX";
    }
    leaf gateway-algorithm {
        type identityref {
            base ax:distribution-algorithm;
        }
        description
            "The gateway algorithm used by the Neighbor DRNI
            Gateway to assign frames to Gateway Conversation IDs.";
        reference
            "7.4.1.1.41 of IEEE Std 802.1AX";
    }
    leaf gateway-conv-service-digest {
        type binary;
        description
            "The MD5 Digest of the Neighbor DRNI Gateway's
            the Home_Admin_Gateway_Conv_Service_Map. Obtained
            from Gateway_Conv_Service_Digest in the
            Neighbor_Gateway_State TLV last received from the
            Neighbor DRNI System.";
        reference
            "7.4.1.1.42 of IEEE Std 802.1AX";
    }
    container gateway-available-mask {
        description
            "A vector of Boolean values, with one value for each
            possible Gateway Conversation ID. A 1 indicates that
            the Neighbor DRNI Gateway Port is eligible to be
            selected to pass that Gateway Conversation ID, and
            a 0 indicates that it is not eligible.";
        reference
            "7.4.1.1.43 of IEEE Std 802.1AX";
        uses drni-mask;
    }
    container gateway-preference-mask {
        description
            "A vector of Boolean values, with one value for each
            possible Gateway Conversation ID. A 1 indicates that
            the Neighbor DRNI Gateway Port is the preferred
            Gateway when both DRNI Gateways have the Gateway
            Conversation ID enabled in the gateway-available-mask,
            and a 0 indicates that it is not preferred.";
        reference
            "7.4.1.1.44 of IEEE Std 802.1AX";
        uses drni-mask;
    }
}
}
}

augment "/if:interfaces/if:interface/if:statistics"
+ "/dotlax:lag-stats" {
    when '../..//dotlax:lag/dotlax-drni:drni' {
        description
            "Applies to aggregators with DRNI present.";
    }
    description
        "Augment interface statistics with DRNI statistics.";
    container drni-stats {
        description
            "Contains DRNI specific statistics.";
        leaf drcpdus-rx {
            type yang:counter64;
            description
                "The number of valid DRCPDUs received on this
                Intra-Relay Port.";
            reference
                "7.4.1.1.45 of IEEE Std 802.1AX";
        }
        leaf illegal-rx {
            type yang:counter64;
            description

```

```
    "The number of frames received on this Intra-Relay  
    Port that carry the DRCP EtherType value,  
    but contain a badly formed PDU.";  
    reference  
    "7.4.1.1.46, 9.6.1.4 of IEEE Std 802.1AX";  
  }  
  leaf drcpdus-tx {  
    type yang:counter64;  
    description  
    "The number of valid DRCPDUs transmitted on this  
    Intra-Relay Port.";  
    reference  
    "7.4.1.1.47 of IEEE Std 802.1AX";  
  }  
}  
}
```

Annex A

(normative)

Protocol implementation conformance statement (PICS) proforma⁷

Insert the following rows at the end of the table in A.2.1

A.2.1 Major capabilities/options

Item	Feature	Status	References	Support
YANG	Does the implementation support management operations using YANG modules?	O	10.6	Yes [] No []
YANG modules	Is the ieee802-dot1ax-types module supported?	YANG:M	10.7.1	N/A [] Yes []
	Is the ieee802-dot1ax-linkagg module supported?	YANG:M	10.7.2	N/A [] Yes []
	Is the ieee802-dot1ax-drni module supported?	DRNI AND YANG:M	10.7.3	N/A [] Yes []

⁷ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Annex G

(informative)

Bibliography

Insert the following two bibliography entries after [B8] in Annex G:

[B9] IETF RFC 7803, Changing the Registration Policy for the NETCONF Capability URNs Registry, February 2016.

[B10] IETF RFC 8040, RESTCONF Protocol, January 2017.

[B11] OMG Unified Modeling Language (OMG UML), Version 2.5, March 2015.⁸

⁸OMG documents are available from the Object Management Group (<https://www.omg.org/>).